XRGait: Immersive Gait Training Visualization with Integrated Sensing

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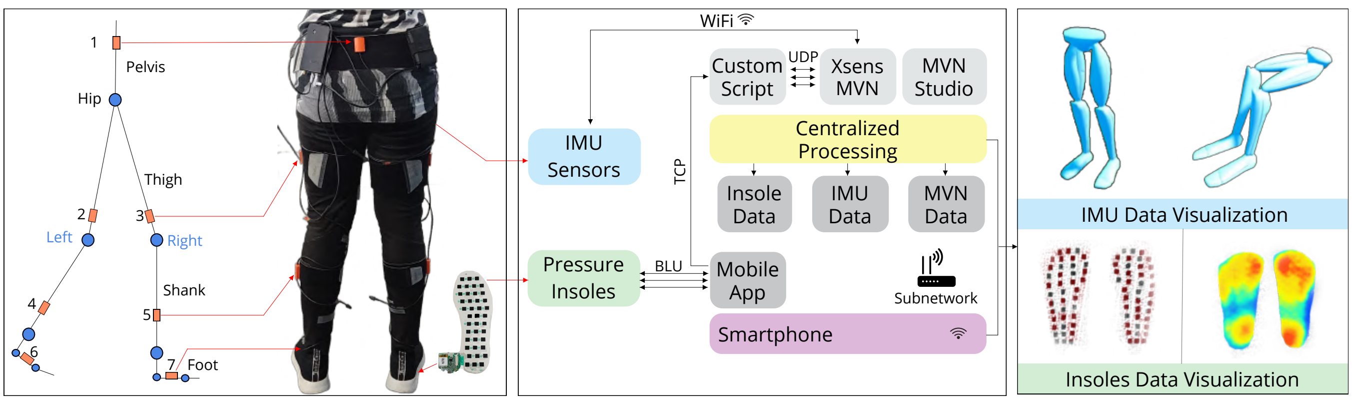

(a) Sensors alignment
(b) Overview of the data flow
(c) Visualisation and feedback

Figure 1: XRGait system overview: (a) Wearable sensors are placed on the patient and calibrated, (b) with the data flow mapped through the main processing components, integrating smart pressure insoles and wearable motion tracking sensors with immersive technologies, (c) to provide the patient with real-time visualization and feedback during gait training sessions.

Abstract

Gait training and rehabilitation have traditionally been done through movement observation during in-person sessions with medical professionals and follow-up exercises at home. However, these methods are costly, have low patient adherence, and have a decreased accuracy in assessing and modeling patient gait. We present XRGait, a system that seamlessly integrates wearable sensors with immersive technologies to provide real-time support for gait training and rehabilitation. Our system combines smart insoles and wearable motion tracking sensors with immersive displays to provide the patient with real-time feedback about their gait within an engaging training environment. We conducted two studies that compared various system environments and gait visualizations to inform optimal system usability and user experience. Our findings demonstrate that the level of immersion is a critical factor for user engagement in biofeedback-driven gait training, with immersive VR significantly outperforming mobile AR and non-immersive screen-based displays and participants found the real-time visualizations and feedback very useful, with a preference for the full lower body visualization. We discuss these findings, along with limitations and recommendations for future work.

CCS Concepts

• **Human-centered computing → Human computer interaction (HCI)**; *Usability testing*.

Keywords

Gait training, wearable sensors, virtual reality, mixed reality

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1 Introduction

Gait analysis can be defined as the observation of one's walking pattern and any associated deviation from one's regular walking pattern. Alongside this, gait training is a means of rehabilitation, typically prescribed by a doctor or physiotherapist. Both gait analysis and gait training are often used in conjunction for a variety of purposes. These include a form of physical therapy for post-trauma or post-stroke patients; to improve locomotion, performance, and form for athletes; and to help patients with balance disorders. Gait training and rehabilitation have traditionally been performed through movement observation during in-person sessions with healthcare professionals and follow-up exercises at home. However, these methods are costly, have low patient adherence, and have a decreased accuracy in assessing and modeling the client's gait when training at home.

The developments in tracking and wearable technologies, such as smart insoles have recently allowed not only to improve posthumous analysis and monitor progress, but also to provide real time feedback [12, 18]. Similarly, the recent pervasiveness and affordability of immersive technologies (AR, VR) for the everyday consumer and the already documented benefits of using this technology in rehabilitation have opened up new opportunities [11, 23]. Our work aims to leverage these recent technological developments and explore solutions that complement current gait training practices by developing ways to practice in at-home or health settings with enhanced engagement and adherence. Based on theories of immersion and presence [34], we hypothesized that the more immersive VR condition would lead to higher engagement and usability compared to the mobile AR condition.

We present XRGait, a system that integrates wearable sensors with immersive technologies to provide real-time support and enhanced patient engagement for gait training. The development of the XRGait system followed an iterative user-centered approach with a user study conducted after each main iteration. In the first iteration, we used the PediSol smart insoles sensors, developed two visualizations (SensorDot and HeatMap) and implemented the system in two modalities: a mobile AR application and a VR application. Feedback from user study and health experts informed the second implementation, where we used motion caption sensors (Xsens MVN) to create a more comprehensive gait model and visualized users' lower body movements in two modalities: VR and non-VR (large screen display).

The main contributions of this research are:

(1) Novel Sensing Technologies for Gait Visualization: We present XRGait, a gait training visualization system that seamlessly integrates advanced sensing technologies, including Pressure Insoles and IMU sensors, to provide real-time visual feedback.

(2) Gait to Real-time Visual Feedback Transformation: We evaluated two types of real-time visual feedback based on insole sensor data (SensorDot and HeatMap Visualization) and lower-body motion from IMU sensor data, delivered across mobile AR, VR, and non-VR platforms.

(3) Comparative User Evaluation of Mobile AR vs. VR vs. non-VR: We compared the usability, user preference, and engagement of mobile AR, VR, and non-VR modalities for gait training.

2 Related Work

The advancement of technology in rehabilitation has led to significant developments in wearable sensor-based gait monitoring and feedback. In this section, we examine these advanced methods and their impact on gait analysis and rehabilitation.

2.1 Wearable Sensor-Based Gait Monitoring

Wearable sensor-based insoles have emerged as a promising alternative for gait re-training that is both portable and user-friendly [40]. These insoles monitor foot-ground interactions through multiple pressure sensors to assess gait dynamics [1]. Development of such tailored insoles offers real-time feedback for continuous gait improvement and is especially valuable for home-based monitoring due to their portability and ease of use [27]. Integrating multiple sensors within insoles, such as eTextile-based pressure sensors and an inertial measurement unit, has shown to overcome the limitations of traditional gait analysis by providing comprehensive gait data [22]. Building on this approach, the Xsens DOT wearable sensor platform advances wearable technology by combining IMUs and pressure insoles to collect kinematic and kinetic data during movement [2]. It supports wireless streaming for real-time analysis and feedback opening up the potential to be used beyond traditional lab settings. Such wearable sensor-based systems have shown consistent performance across lab studies. For example, Gorton et al. [14] found significant kinematic measurement differences between motion analysis labs, highlighting the need for wearable sensors that can deliver consistent data in both clinical and home environments.

Gait and balance monitoring using wearable sensors has been explored also in home-based training programs. Insole pressure sensors have been shown to assess post-stroke gait without affecting the gait itself, indicating their potential as reliable clinical endpoints even in home environments [24]. Wearable sensors proved effective for gait monitoring [29], despite no significant changes in balance variables over time. The study also found that while smartphones are suitable for measuring different gait parameters, they are less effective for balance assessments. This finding supports the use of dedicated wearable sensors for more accurate and reliable balance monitoring. In addition to monitoring, wearable sensors have been utilized to enhance gait and balance training, demonstrating improved balance more effectively than traditional exercises, which is crucial for patients with balance disorders [13]. Insole-based systems are also useful for health monitoring, offering portability and ease of use in home settings [36].

Despite these advancements, there are challenges associated with wearable sensor technology, including the need for further validation, the integration of complex data into user-friendly formats, and ensuring long-term engagement with users.

2.2 Real-Time Feedback in Gait Assessments

The use of real-time feedback in gait assessments has been a focus in recent rehabilitation research, particularly in enhancing gait symmetry and improving overall gait mechanics [30]; with various studies exploring different aspects of real-time feedback for effective rehabilitation. Caprani et al. [7] and Stoutz et al. [35] highlighted the importance of designing systems that are both user-friendly and aligned with the needs of healthcare professionals. These studies emphasized the necessity for clear, actionable feedback and intuitive interfaces, both critical for user engagement and the effective application of such technologies in real-world settings.

Russell et al. [30] studied the impact of visual feedback on reducing the metabolic demand during gait retraining. This study confirmed the effectiveness of visual feedback but it did not fully explore how different types of feedback could impact training outcomes across various user demographics.

Sangani et al. [31] demonstrated the potential of using avatar-based feedback in VR environments to improve gait symmetry, particularly for stroke patients. While this work highlighted the benefits of visual feedback, it did not incorporate sensing technologies into the feedback loop. Seals et al. [33] proposed design recommendations for integrating sensor data into clinical practice, emphasizing the importance of ease of use and actionable feedback, However, their work primarily identified challenges rather than proposing specific solutions. Khan et al. [18] and Arens et al. [4] focused on the technical and usability requirements for effective gait analysis which laid the groundwork for developing VR-based gait analysis tools, yet there remains a need for systems that provide continuous, adaptive feedback suitable for everyday use.

Anson et al. [3] investigated the effects of visual feedback on trunk control, a critical aspect of maintaining balance during gait. While their work contributes to understanding the role of visual feedback in postural control, it does not address the integration of such feedback into real-time monitoring systems. van Gelder et al. [38] emphasized the role of biofeedback in correcting abnormal gait mechanics. However, these studies primarily concentrated on clinical applications of biofeedback, leaving a gap in exploring effective implementation in home-based or unsupervised settings.

XRGait builds on these foundational studies by providing real-time feedback to users alongside integrating novel sensing technologies for continuous gait monitoring outside clinical settings.

2.3 Integration of AR and VR-Based Gait Training

The integration of Augmented Reality (AR) and Virtual Reality (VR) technologies into gait training has been extensively explored, particularly in stroke rehabilitation and gait training for older adults and children with cerebral palsy [6]. Research has demonstrated the potential of both technologies to improve key gait metrics. For instance, AR-based systems have been shown to enhance postural control and improve variables like walking velocity and stride length, though these studies are predominantly situated in controlled clinical environments [20, 25]. Similarly, VR training has proven effective in significantly improving walking capability, gait symmetry, and community ambulation, often surpassing or offering a viable alternative to traditional rehabilitation methods [8, 9, 11, 21, 41]. The benefits of immersive VR are further evidenced by increases in walking speed, overall performance [39], walking distance, and physical fitness [10].

However, prior work has two key limitations. First, many studies focus on controlled clinical settings [16, 20], with limited exploration of home-based, unsupervised rehabilitation. Second, many VR interventions rely on generic game-based environments [8] rather than visualizations informed by real-time gait measurements from integrated wearable sensors.

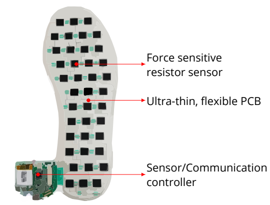


Figure 2: The PediSol Smart Insole used in the XRGait system. This is an insole inserted into the user's shoe with a grid of pressure sensors to track how the user balances their weight during gait training.

Our work, XRGait, builds on this foundation but addresses these gaps by extending AR and VR into home-based training programs, moving beyond prior research by: (i) developing a system to provide real-time feedback on gait training using integrated wearable sensors, moving beyond generic game-based VR; (ii) directly comparing multiple consumer-grade modalities (Mobile AR, Immersive VR, and Desktop display) to inform the design of practical and accessible solutions; and (iii) evaluating these systems with a focus on user experience, engagement, and usability—critical factors for long-term adherence in unsupervised, home-based settings. Through this approach, we aim to bridge the gap between clinically proven efficacy and real-world, everyday application.

3 XRGait Implementation

The development of the XRGait system followed an iterative approach with a user study conducted after each main iteration. In the first iteration, PediSol smart insoles were used to create SensorDot and HeatMap visualizations in mobile AR and VR environments. Based on user feedback, the next iteration employed IMU Sensors (Xsens MVN Link [1]) and visualized users' lower body movements in two different environments: VR and non-VR (large screen display). The gait training exercises were chosen in consultation with clinicians and had difficulty variability within each exercise so they are suitable for any participant (Figure 3a and Figure 6a). The clinicians also provided feedback on the first iteration of the system which led to implementing a lower body visualisation over the sole-only.

XRGait was implemented using the Unity game engine (version 2020.1.16f1) and ran on a machine with an Intel Xeon W-2133 3.60GHz CPU, 16GB of RAM, and a GeForce RTX 2080 Ti GPU. This version of Unity was chosen for its compatibility with AR Foundation, a cross-platform framework that supports AR development for both Android and iOS devices. A Samsung Galaxy S10 mobile phone was used to run the AR application. AR functionality was achieved using AR Foundation and the ARCore XR Plugin. The Meta Quest 2 VR headset was used for the VR condition, providing an immersive virtual environment. The virtual training environment consisted of 3D assets of a straight walking path through a forested area.

[1] https://www.movella.com/products/motion-capture/xsens-mvn-link

3.1 System Design

A key motivation for the design and development of XRGait was to provide a system that can visualise gait information accurately and in real time, while being simple to use by lay people at home, keeping them motivated in continuing their exercises between clinical visits. Great care was taken in designing the overall gait training experience so it is easy to understand, engaging, and actionable for both users and clinicians for purposes like real-time corrections, personalized training and rehabilitation. Across all display modalities (mobile AR, VR, non-VR) we created a safe environment for users (e.g. setting a boundary for the VR, reminders to prompt users to be aware of their surroundings) and provided on-boarding and clear instructions to the exercises.

We implemented two types of sensors (i) Smart Insoles (40 pressure sensors) and (ii) Xsens MVN Link (seven sensors positioned in specific locations of the lower body) to accurately capture the motion and orientation during gait analysis. Extrinsic calibration was required to ensure the sensors are correctly aligned with the appropriate body segments (Figure 1a).

3.1.1 ***Pressure Insoles (PediSol Smart Insoles):*** XRGait utilizes two PediSol smart insoles, each equipped with a wireless communication controller capable of transmitting data in real-time. As shown in Figure 2, these insoles contain 40 force-sensitive resistors that detect variations in physical force, pressure, and weight across the foot while the user is walking or standing.

The controller transmits the data from each sensor to an app on the tablet using Bluetooth Low Energy (BLE) which is then forwarded to the Unity application, either on the desktop (for VR) or the mobile phone (for AR). If pressure is detected on a given sensor (i.e., the integer returns a value greater than zero), the corresponding sensor will turn a shade of red with the intensity of the red color based on the amount of pressure applied.

3.1.2 ***IMU Sensors (Xsens MVN Link):*** Inertial Measurement Units (IMUs) measure acceleration, orientation, and rotational velocity, capturing detailed kinematic data related to body movement. We used the Xsens MVN Link for motion capture, which is equipped with seventeen sensors and is designed to capture high-impact and highly dynamic movements and can track multiple subjects within a 150-meter wireless range. Since our system for gait analysis focuses only on the lower body, it uses seven sensors placed on the feet, shins, thighs, and pelvis to capture detailed motion data of the lower body. The software platform Xsens MVN Link is used to process data from Xsens IMU sensors for motion analysis. IMU data is transmitted to the computer via UDP, which is then processed in tandem with MVN Studio to produce accurate motion visualizations for real-time processing and analysis.

MVN Studio passes the processed data directly into Unity where it is used to generate real-time visual representations. This was achieved by integrating motion capture data from IMU sensors with a rigged 3D leg model created in Blender; the lower body was prioritized in the visualization as this approach allows for more precise tracking of key joints and simplifies the setup. By streaming the data into Unity, users can see their movements replicated in a virtual 3D environment. This real-time visualization is crucial for providing immediate feedback, helping users adjust their movements during gait analysis or rehabilitation sessions. For example, if the user's posture is incorrect or their gait is uneven, these issues will be visible in the 3D visualization.

4 XRGait Evaluation

We conducted two user studies and an expert consultation to determine optimal system usability and enhance user experience by comparing various system environments (immersive and non-immersive settings), sensor inputs (Insoles and Xsens) and gait visualizations:

Study 1 - **XRGait in mobile AR and VR**: This study evaluated the first implementation of XRGait that used sensor input data from the PediSol Smart Insoles and compared the user experience two gait visualization methods (Heatmap and SensorDot) between mobile augmented reality (AR) and virtual reality (VR) environments.

Study 2 - **XRGait in VR and Non-VR**: This second study used sensor input data from the Xsens visualised as a 3D model of the lower body using immersive (VR) and non-immersive (desktop) displays to assess user experience and preference.

4.1 Study 1: XRGait in mobile AR and VR

The goal of this study was to determine the most effective method for displaying the gait data captured by the PediSol smart insoles to users, ensuring a rich and user-friendly experience. We compared the effectiveness of presenting this data in VR versus overlaying it on the real world using mobile AR. Through incorporating feedback from health professionals and existing work, two visualization methods were chosen for implementation (see Figure 1c):

- **SensorDot Visualization**: represents foot pressure using individual dots or markers on a foot map. Each dot corresponds to a pressure sensor, and its color intensity indicates the magnitude of pressure at that location.
- **HeatMap Visualization**: displays foot pressure distribution using a color gradient. Areas of higher pressure are represented by warmer colors like red or orange, while areas of lower pressure appear in cooler colors like blue or green.

4.1.1 Experimental Design. We had the following hypotheses:

(H1): HeatMap visualisation will be rated significantly higher for usability than the SensorDot visualisation, and will be significantly preferred by users.

(H2): The mobile AR will be rated significantly higher than the VR in both usability and user preference metrics.

We employed a 2×2 within-subjects study design with four conditions:

Participants experienced two conditions each: one in VR (with SensorDot or HeatMap visualisation) and one in AR (with the other visualisation). The order of conditions was randomized with each participant experiencing one out of possible four combinations:

- VR with HeatMap followed by AR with SensorDot
- VR with SensorDot followed by AR with HeatMap
- AR with HeatMap followed by VR with SensorDot
- AR with SensorDot followed by VR with HeatMap

Table 1: Our four study conditions using a 2x2 within-subjects design

	HeatMap	SensorDot
VR	VR with HeatMap Visualisation	VR with SensorDot Visualisation
AR	AR with HeatMap Visualisation	AR with SensorDot Visualisation

Each participant completed ten minutes of exercises for each environment (VR/AR) with the associated visualisation (SensorDot/HeatMap) followed by a usability questionnaire. Upon completion of the two sessions, a post-study interview was conducted to gather insights into participants' experience.

4.1.2 Participants. We recruited fourteen participants (seven female, seven male) through university mailing lists and flyers posted around the campus. The age range of participants was from 21 to 34 years (M = 26.4, SD = 4.6). All participants had normal or corrected-to-normal vision. Most participants had limited experience with VR/AR technologies, with usage reported as follows: rarely (n = 6), weekly or more (n = 2), monthly or more (n = 3), annually or more (n = 3), and never (n = 1). None of the participants reported any prior experiences of simulator sickness, significant motion sickness, or any related illnesses such as epilepsy.

4.1.3 Measures. We used a mixed-methods design, utilizing both quantitative and qualitative data gathering methods to obtain understanding of user experiences and preferences with the implemented system.

Demographic Information: Participants completed a demographic questionnaire collecting background information, including age, gender, foot size, prior experience with VR, familiarity with gait training exercises, and exposure to physiotherapy.

System Usability: The System Usability Scale (SUS) [5] questionnaire was used after each session to to collect participants' opinions about the perceived usability of the XRGait system, training exercises, and visualisations in both mobile AR and VR.

Semi-Structured Interview: A post-study semi-structured interview was conducted to gather participants' feedback on their experience and the effectiveness of the proposed system. Participants' responses during the interview were deductively coded and grouped into categories of interest using thematic analysis. To quantify the emotional tone of participant responses, lexicon-based sentiment analysis was used on a scale from -1 to +1, with +1 representing extremely positive sentiments and -1 representing extremely negative sentiments [37]. To identify underlying dimensions in user feedback related to usability and engagement, exploratory factor analysis was employed [26]. Finally, to further contextualise participants' interpretation, chi-square test was used to examine the relationship between categorical variables e.g. system type (VR vs AR) and usability sentiment (positive vs negative).

4.1.4 Procedure. Participants were first provided with an information sheet about the study and upon consent, completed a demographic questionnaire. Participants then were given the shoes with the PediSol Smart insoles to wear and a brief calibration was performed to ensure accurate tracking and data collection during the exercises.

In the AR condition, an Android AR app was utilized to map the physical space near the participants' feet, and showed the assigned gait visualization. For the VR condition, a virtual nature setting was created, allowing participants to move freely within the assigned boundary. Participants experienced the same set of gait training exercises in both conditions, each using one out of the two available ways to visualize the gait - HeatMap or SensorDot visualization. Participants initiated the test by pressing a start button within their virtual environment (see Figure 3c) or on the mobile phone screen. A sequence of exercises then appeared in front of them (see Figure 3a), These exercises lasted for ten minutes, after which participants completed the SUS questionnaire to evaluate their experience. This was repeated for both conditions.

At the end of the session, a brief audio-recorded interview was conducted with each participant to collect qualitative feedback. The entire procedure took about 45 minutes for each participant. As a token of appreciation of their contribution, all participants were entered into a draw to win one of two $50 coupons.

4.1.5 Quantitative Results. The quantitative results show a significant preference for the VR modality, which demonstrated higher system usability and more positive user sentiment overall.

System Usability: A Shapiro-Wilk test found the SUS scores to be normally distributed ($p = .81$). A repeated-measures ANOVA found a significant difference in SUS scores between display types ($F(1, 24) = 5.49$, $p = .03$), with VR ($M = 77.30$, $SD = 15.20$) scoring significantly higher than AR ($M = 64.30, SD = 13.30$). No significant difference was found between the SensorDot and HeatMap visualisations ($F(1, 24) = 0.37$, $p = .55$), and no interaction effect was observed between display type and visualisation ($F(1, 24) = 0.13$, $p = .73$). See Figure 4 for a graphical representation of these results.

Interview Sentiment Analysis: Sentiment analysis of the interviews showed the HeatMap visualization in VR received a strongly positive sentiment score (**+0.85**) due to its immersive qualities, clarity, and ease of understanding (see Figure 5). SensorDot in VR also scored positively (**+0.72**) due to its immersive nature, which made exercises feel more like a game, enhancing engagement. In contrast, SensorDot in AR received a negative sentiment score (**-0.35**), as participants found it bit cumbersome and unintuitive. HeatMap in AR visualization received a neutral sentiment score (**+0.05**); while participants acknowledged the clarity of the HeatMap and the AR features, the overall sentiment remained neutral.

Preference for Display Modalities vs Visualization: There was a statistically significant preference for VR over AR in terms of usability ($\chi^2 = 6.84$, $p < .01$). A significant preference for the HeatMap visualization over the SensorDot was also found ($\chi^2 = 9.24$, $p < .005$). These results indicate that VR combined with HeatMap visualization is preferred for gait training exercises, highlighting the importance of clarity and immersion in such tools.

Key Components Driving Preference: Factor analysis identified two main usability dimensions: ***Ease of Setup*** (higher for AR) and ***Physical Interaction*** (higher for VR). For engagement, two key dimensions emerged: Immersive Enjoyment (favoring VR) and

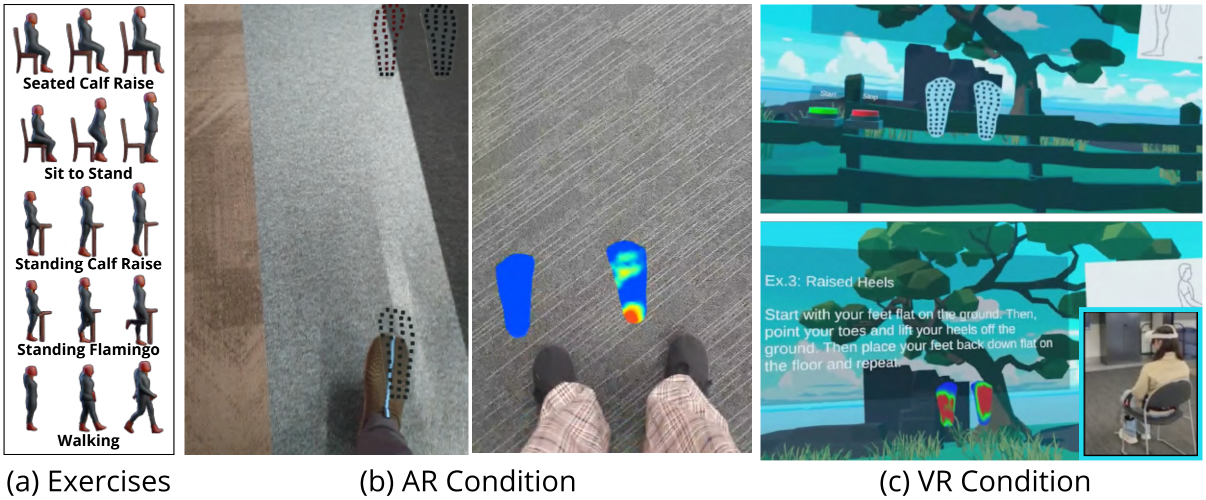


Figure 3: An overview of the first user study. (a): The exercises completed by participants in each condition. (b): The AR interface with SensorDot (left) and HeatMap (right) visualisations. (c): The VR interface with SensorDot (top) and HeatMap (bottom) visualisations.

Repetitive Task Fatigue (indicating mobile AR was less engaging). The HeatMap scored higher on clarity, making it easier for participants to interpret in real-time. In contrast, the SensorDot visualization had lower clarity and usability scores, requiring more effort to interpret.

4.1.6 Qualitative Feedback. Thematic analysis of the interviews reveals three key themes: usability of VR and AR environments, visualization preferences, and impact on exercise performance.

Usability of VR and AR Environments: Participants generally preferred the VR environment for its immersive, engaging, and ease of use during exercises and mentioned in particular appreciating how the design of the environment made it easy and clear to understand what to do. The AR was viewed as more accessible and easier to set up for regular use at home, suggesting that the AR option may be more suited for users looking for convenience. Overall though, the AR environment was seen as more cumbersome as it required holding the phone and looking at a small screen which was challenging to juggle while performing the exercises and focusing on their gait, thus rejecting hypothesis **H2**.

In contrast, the VR environment allowed participants to focus solely on the exercises without any external distractions or having to hold an additional device.

> *"I felt the VR was easier because, (with) the AR, I was worried about the phone and what happening in the background"* — (P6; Study 1)

Visualization Preferences: Participants consistently emphasized the clarity of the HeatMap visualization over the SensorDot, as many found it to be more intuitive and easier to understand and interpret pressure distribution results in real-time, partially supporting **H1**.

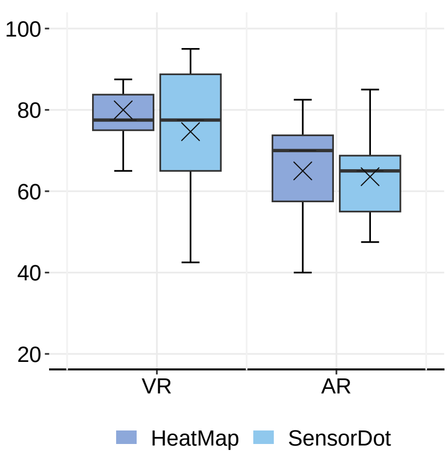


Figure 4: The results of the SUS by display type and visualisation.

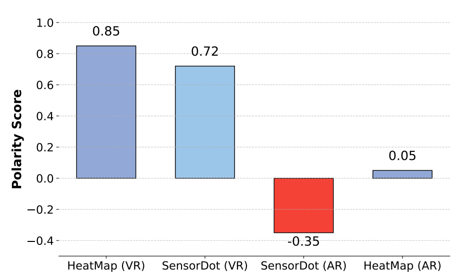


Figure 5: Sentiment Analysis of Visualizations in VR and mobile AR

> *"The HeatMap was easier for me to be like, oh, that's what's happening, like when I put pressure on my heel"* — (P1; Study 1)

In contrast, the SensorDot visualization was perceived as harder to interpret, with several participants noting that it required more cognitive effort, even though it was appreciated that it provided more detailed information.

> *"The dots were not as clear, and I didn't understand them as quickly"* — (P6; Study 1)

All participants discussed the benefits of real-time, actionable feedback during physiotherapy exercises, which the HeatMap seemed to support more effectively, and some suggested that having a more embodied visualisation (full or lower body) can provide more comprehensive gait feedback.

Impact on Exercise Performance: Participants highlighted that the VR environment increased engagement and made the exercise experience more enjoyable compared to AR.

> *"The VR one was fun,it was, kinda like a game... but the idea is like it's for like physio or whatever so it's like makes it more interesting ...much better in comparison to the AR where it's less like that"* — (P13; Study 1)

The visualisation had a mixed impact on their exercise performance. While the HeatMap helped with body awareness and posture correction, some participants felt that the visual feedback was not always necessary for performing exercises.

> *"I wasn't really looking at the phone as much as I thought I would because since it was the same thing on the phone and in my environment, I just kinda, like, looked at the actual floor instead of the floor on my phone"* — (P12; Study 1)

Due to the repetitive nature of the exercises several participants suggested incorporating more interactive or gamified elements into both the AR and VR systems to enhance engagement.

> *"If I could have some music or something like that. So, like, maybe it would be easier something to, like like, give some motivation"* — (P4; Study 1)

> *" ...some fake world stuff onto an app would be quite fun. Like, gamifying the hair... because I think it's more fun to do exercises if it's, like, visually more interesting. It's like Pokemon Go in a way"* — (P12; Study 1)

Furthermore, some participants, pointed out that the immersive VR environment made the exercises feel less tedious and suggested that being able to personalise or integrate this system with existing health apps would make it more appealing and improve adherence to physiotherapy regimens, especially for long-term recovery.

> *"I can definitely see (the VR system) being used together with some of the other health apps or running apps"* — (P1; Study 1)

4.2 Study 2: XRGait in VR and non-VR

The results from Study 1 and the expert consultation informed the next iteration of the XRGait design and implementation. Given the VR display modality was the preferred and most engaging, we

Table 2: Summary of Hypotheses and Key Findings from Study 1

	Supported?	Key Findings
H1	**Partially**	HeatMap was significantly preferred over SensorDot (χ^2 = 9.24, p < .005), but there was no significant difference in usability (SUS: p = .55).
H2	**Not Supported**	VR scored significantly higher than mobile AR in both usability (SUS: p = .03) and user preference (χ^2 = 6.84, p < .01). Sentiment also favored VR.

refined the design of the VR environment and opted to compare with an non-VR (large screen display), instead of the AR, as an alternative low cost, accessible option. We further implemented, based on the expert consultation, a more comprehensive approach with regards to the sensors and visualization, utilizing the Xsens motion capture sensors to track and visualize participants' lower body movements during gait training exercises in both VR and non-VR environments. This second study aimed to explore users' preference for the training environment and its impact on the sense of presence, embodiment, and overall effectiveness in performing gait training exercises.

4.2.1 Experimental Design. For this study, we were interested in the following hypotheses:

- **(H3):** The VR environment will lead to higher participant engagement, embodiment, and efficacy levels compared to the non-VR environment.
- **(H4):** The VR environment will involve increased task workload compared to the non-VR.

Since this study involves two display modalities (VR and non-VR) and a single visualization type (lower body tracking), a 2×1 within-subjects design was employed with two conditions.

(Exploratory H∗): Though not one of our formal hypotheses, we were also interested in how these visualisations can affect participants' understanding of their gait and posture, and how this awareness might be applied to future clinical trials; we explore these aspects further in our qualitative results.

4.2.2 Participants. We recruited 20 participants (13 females, 7 males) from 20 to 66 years of age (M = 28.6, SD = 9.34). A subset of the participants had taken part in the first study. The majority had prior experience with VR systems, while a smaller group (N = 4) had never used a VR system before. Participants were recruited through advertisement flyers distributed across the campus and via word-of-mouth referrals from volunteers. None of the participants reported any prior experiences of simulator sickness, significant motion sickness, or any related illnesses such as epilepsy.

4.2.3 Measures. Similar to the first study, we utilized both quantitative and qualitative data-gathering methods. We collected the same demographic information as before, along with the following measures:

Task Workload: Participants completed the NASA-TLX [15] questionnaire after each condition to evaluate the cognitive and physical demands of the gait training exercises in both conditions.

(a) Exercises (b) VR Condition (c) non-VR Condition

Figure 6: Overview of the Study 2: Participants perform (a) a set of exercises under two conditions: (b) within a virtual reality environment with real-time feedback (VR Condition) and (c) while viewing the environment on a screen without VR immersion (non-VR Condition).

NASA-TLX was measured across all six dimensions: mental, physical and temporal demand, performance, effort, and frustration. NASA-TLX was administered using a 7-point Likert scale for each subscale for its simplicity and ease of use for participants.

Presence and Embodiment: The sense of presence was measured using the General Presence and Spatial Presence subscales of the *iGroup Presence Questionnaire* (IPQ [32]). For embodiment, we incorporated several key components to assess how users perceive and interact with virtual bodies [19]. This questionnaire included items that asked participants to rate their feelings of control, body ownership, and the integration of their virtual body with their real body movements.

Semi-Structured Interview: A post-study semi-structured interview, similar to the first study, was conducted to gather insights into participants' experiences, the training methods' effectiveness, and the benefits and drawbacks of each environment and visualization. Interview and open-ended questionnaire responses were deductively coded and categorized using thematic analysis.

4.2.4 Procedure. Upon arrival, participants were briefed, signed a consent form, and completed a demographic questionnaire. They were then fitted into the Xsens MVN Link sensors and calibrated for height and foot size and joint alignment. Participants performed the gait training exercises (see Figure 6) through experiencing the VR and non-VR (large screen) conditions in a randomized order using a balanced Latin Square design. After each condition, they completed questionnaires on task workload, presence, and embodiment. Following both conditions, a short semi-structured interview was conducted to discuss their experiences, preferences, and ease of performing exercises. As a token of appreciation, participants were entered into a draw for three $50 coupons, with three winners randomly selected.

4.2.5 Quantitative Results. **Sense of Presence:** Participants' sense of presence results are shown in Figure 7a. Since the data violated normality assumptions based on the Shapiro-Wilk test ($p_{PRES} < .01$, $p_{SP} < .0001$), the Wilcoxon Signed-Rank Test (WSR) was conducted with Holm-Bonferroni correction to assess the significance of differences between these conditions. The WSR revealed a significant difference in General Presence ($W = 339.5$, $p < 0.001$) between the VR ($Med = 5.10$, $IQR = 2.50$) and non-VR ($Med = 2.00$, $IQR = 1.50$) conditions, suggesting that participants experienced a significantly higher sense of general presence in the VR environment compared to the non-VR environment. For Spatial Presence, the WSR also indicated a significant difference ($W = 1285.5$, $p < 0.001$) between the VR ($Med = 6.00$, $IQR = 2.25$) and non-VR ($M = 2.00$, $IQR = 2.25$) conditions. Our findings suggest that participants experienced a higher sense of spatial presence in the VR compared to the non-VR environment.

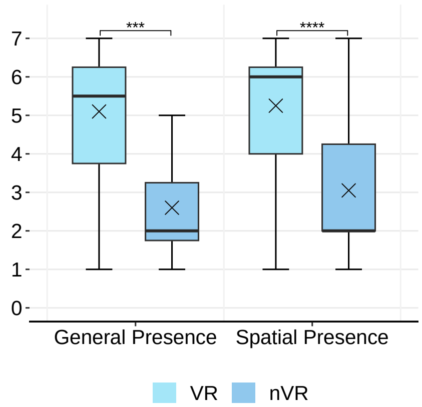
Embodiment: The embodiment results across four dimensions are shown in Figure 8. *Agency* was found to be non-normal ($p = .01e^{-5}$), and a WSR revealed no significant difference ($W = 969.0$, $p = 0.096$) between the VR ($Med = 6.00$, $IQR = 2.00$) and non-VR ($Med = 5.00$, $IQR = 2.25$) conditions.

Ownership was similarly found to be non-normal ($p < .01$), and a WSR found no significant difference ($W = 259.5$, $p = 0.10$) between the VR ($Med = 5.00$, $IQR = 1.25$) and non-VR ($Med = 3.00$, $IQR = 4.00$) conditions.

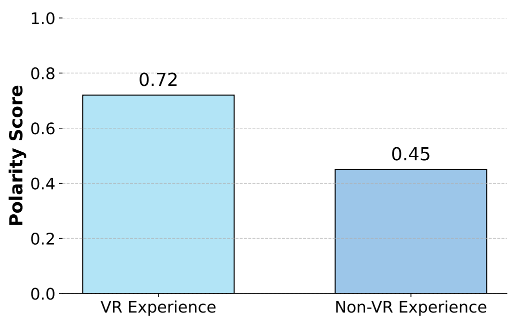
Similarly, *Self-location* was non-normal ($p = .03$), and a WSR did not find a significant difference ($W = 267.5$, $p = 0.07$) between the VR ($Med = 5.00$, $IQR = 2.00$) and non-VR ($Med = 4.00$, $IQR = 3.00$) conditions.

Perception of More Bodies was likewise non-normal ($p = .01$), however a WSR did find a significant difference ($W = 292.0$, $p = 0.012$) between the VR ($Med = 5.00$, $IQR = 2.25$) and non-VR ($Med = 3.00$, $IQR = 3.00$) conditions. This suggests that participants felt a stronger presence or awareness of other bodies in the VR environment compared to a non-VR one.

Task Workload Assessment: The NASA-TLX results across six dimensions are shown in Figure 9. Each participant rated their experience on 7-point Likert scale, with higher scores indicating a greater task load. The WSR with Holm-Bonferroni correction indicated no statistically significant differences between the conditions. P-values for all dimensions exceed the significance threshold



(a) The General Presence (left) and Spatial Presence (right) scores from our second user study.



(b) Sentiment Analysis: VR vs Non-VR Experience

Figure 7: Presence and sentiment analysis results across conditions.

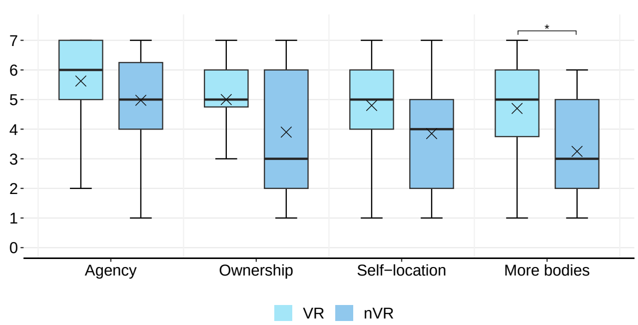


Figure 8: Embodiment questionnaire results from Study 2

($\alpha < 0.05$), indicating no significant difference in task load between VR and non-VR experiences, thus rejecting hypothesis **H4**.

Interview Sentiment Analysis: Sentiment analysis of the interviews showed an overall positive sentiment toward the VR experience compared to the non-VR alternative. Using a polarity score system, with +1 being extremely positive and -1 being extremely negative. *VR Sentiment*: **+0.72** (indicating generally positive experiences), *non-VR Sentiment*: **+0.45** (indicating moderately positive experiences) (see Figure 7b). Participants frequently used words such as 'enjoyable,' 'immersive,' and 'novel' when describing the VR experience, suggesting high engagement and enjoyment levels. Negative sentiment was more concentrated around the words 'overwhelming,' 'dizzy,' and 'confusing,' indicating challenges in instruction clarity.

Preference for VR vs non-VR: To analyze the preference for VR over non-VR, a chi-square test was conducted. This result indicates ($\chi^2 = 7.25$, $p < 0.05$) a statistically significant preference for VR over non-VR, confirming that most participants favored the VR environment. The effect size ($V = 0.51$) suggests a moderate to large association between VR use and participants' preference, emphasizing VR's appeal as the preferred mode for exercises.

Key Components Driving Preference: A factor analysis was conducted to determine the underlying components influencing participants' preference for VR or non-VR environments. The analysis identified three key factors:
Engagement and Immersion Participants who rated high on engagement and immersion consistently preferred VR, suggesting that these are the primary drivers of preference.
Ease of Use and Familiarity Participants who valued ease of use and familiarity tended to prefer the non-VR experience, emphasized simplicity and the lack of distractions in non-VR settings.
Physical Challenge Participants who enjoyed a more physically challenging experience were more inclined to favor VR, which provided a stimulating and demanding environment.

4.2.6 Qualitative Feedback. The key findings from the interview data analysis touched on the use of VR and Non-VR platforms for physical exercise, particularly in the context of movement tracking and feedback.

Effectiveness of the Lower Body Visualization: Overall, participants found the visualization beneficial; in particular those who were less familiar with physiotherapy or structured exercise programs. In the VR mode, the majority of participants (70%) indicated that the visualization provided better feedback on movement accuracy, helping them correct their posture and improve performance:

> *"The visualizer helped me understand how my body was moving and where I needed to make corrections. So it was quite useful."* — (P9; Study 2)

> *"I liked being able to see my movements; it made it easier to correct my form, especially since I haven't done much physiotherapy before"* — (P20; Study 2)

Some participants also highlighted the novelty of seeing their movements displayed visually, describing it as '*surreal*'. However, there were moments of disconnect, where the participants felt that the visualization did not perfectly align with their actual body movements, which led to slight confusion.

> *"yeah I think the walking around in the paddock... if the body was like nearer me... I didn't recognize it as my own body"* — (P18; Study 2)

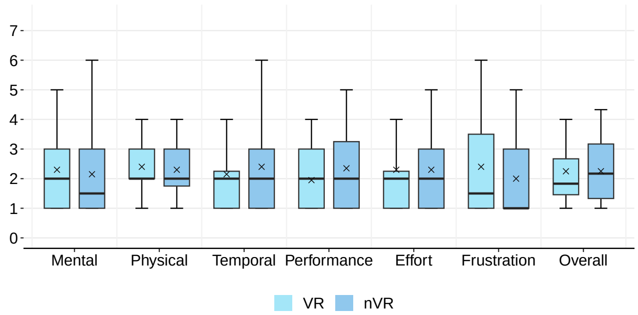


Figure 9: TLX results from Study 2 (lower is better)

When performing familiar exercises, having the visualisation was found useful in both modes which suggests visual aids helped reinforce understanding. However, its impact was not as profound in the non-VR compared to VR; with participants often feeling that they could achieve similar results by simply relying on their body awareness or using a mirror.

Engagement in the VR vs non-VR: The VR mode was found to increase enjoyment and engagement, with participants reporting that the dynamic and immersive environment kept them motivated to complete the exercises. The VR mode was considered easier to customise and added a playful level of difficulty, particularly in exercises requiring balance and activities like walking:

> *"It was more challenging but in a fun way, like you had to really think about your balance"* — (P6; Study 2)

Some participants felt that VR offered better visual instructions, allowing them to follow along with animations more intuitively. Others, however, found the instructions in VR to be less clear due to distractions in the environment or unclear textual descriptions, with some preferring the instructions in audio form. Similarly, a small subset of participants expressed discomfort, such as dizziness or feeling disoriented during VR exercises compared to using the non-VR setting. This suggests a mixed outcome for hypothesis **H3**, as embodiment and efficacy levels in VR were not statistically significant, thus user feedback further indicates partial rejection of **H3**.

Areas for Improvement: Both participants in VR and non-VR suggested several improvements to the visualization and interface. The most common suggestion was more detailed and real-time feedback. They suggested features like muscle activation monitoring or progress indicators to make the experience more informative and rewarding. Other suggestions were about customisation such as being able to change the visual representation of the body, the environment, or add gamified elements.

> *"If I could see which muscles were being worked in real-time, what muscles should be working it would help me focus on improving (...) It would be nice if I could change the setting, maybe even add some fun rewards, like sparkles when I get a movement right"* — (P3; Study 2)

Table 3: Summary of Hypotheses and Key Findings from Study 2

	Supported?	Key Findings
H3	Partially	VR showed significantly higher presence (general and spatial). Only one embodiment subscale was significantly higher (Perception, p = .012). Other embodiment metrics were not significant.
H4	Not Supported	No significant differences in NASA-TLX scores; VR was not perceived as more demanding.
H*	Exploratory Insight	Qualitative responses suggest improved awareness and perceived clinical value in VR, but this was not formally tested.

5 Discussion

This research aimed to leverage the recent developments in immersive reality and sensor technologies to complement the current practice of gait training traditionally performed by healthcare professionals. Our two studies following a user-centered iterative approach aimed to investigate new ways for clients' to practice gait training at home/health settings while providing real-time detailed feedback to enhance their engagement and adherence and achieve their health/rehabilitation goals. In **Study 1**, we used the PediSol smart insoles sensors, developed two visualisations (SensorDot and HeatMap) and implemented the system in two modalities: a mobile AR application and a VR environment. Our **H1** was partially supported: participants significantly preferred the HeatMap visualization over SensorDot, but usability scores did not differ. **H2** was not supported, as VR outperformed mobile AR in both usability and user preference, contrary to our initial expectations that portability and familiarity would favor AR (see Table 2).

Previous work had found several benefits using VR [11, 21, 41] but equally suggested there can be issues e.g. headset may affect balance and posture and proposed the use of AR as an alternative. Our results confirmed the benefits of VR and participants didn't report being negatively affected which may be thanks to hardware improvements (lighter, wireless headsets). In contrast, the AR setup was found more cumbersome and difficult to use concurrently with the gait exercises. Similarly, the clinicians recommended showing a more comprehensive visualisation - full or lower body instead of the sole only - and on a separate (bigger) display instead of holding the phone. The VR was preferred, despite potential purchase and setup costs as it was seen to have engagement and adherence benefits that are more valuable. It was further seen as an opportunity to rent this high-tech equipment to their clients as an addon service plan which can give their practice an edge over competitors.

Based on overall feedback, we then used motion caption sensors as input and visualised users' lower body movements in VR and non-VR (large display). In **Study 2**, we found partial support for **H3**: participants in the VR condition reported significantly higher presence (both general and spatial), and preferred VR overall, indicating higher engagement. However, only one embodiment measure *Perception of Other Bodies* showed a significant difference, while others (agency, ownership, self-location) did not, suggesting that embodiment effects may be more nuanced. **H4** was not supported, as no significant differences were found in task workload, indicating that the added immersion of VR did not increase perceived cognitive or physical demands. Finally, although **H**∗ was exploratory, qualitative feedback suggests that immersive and clear visualizations, particularly in VR, enhanced participants' awareness of their gait and posture, with implications for future clinical applications (see Table 3).

Participants feedback also suggests an *expertise effect* (perceived value of the tool varies depending on the user's prior experience) [17]. Participants who were less experienced with physiotherapy exercises found the heat map useful for improving body awareness. One participant remarked, *"The heat map made me more aware of how I was distributing pressure, which helped me correct my posture."* Conversely, more experienced participants felt that they didn't need the visualizer as much. *"I'm used to using mirrors for feedback, so I didn't rely on the visualizer that much."* This suggests that while the visualizer is beneficial for beginners, its impact may be limited in some cases for those with more experience in physiotherapy.

6 Limitations and Future Work

One of the primary focuses of this paper is on gait, particularly the feet and lower body, as studies have shown that pressure insoles and IMU sensors provide valuable insights into dynamic balance and ambulatory gait, especially for individuals using walkers or canes [28]. However, a full gait analysis must also consider the upper body, including the trunk, arms, and head, which are crucial for maintaining balance, coordinating movement, and compensating for imbalances. The current system also lacks dynamic balance detection, which could be useful for identifying gait deviations and providing valuable stability insights to patients and clinicians.

It is also important to note that this work involved a small study sample consisting of primarily healthy adults. While not including clinical populations was purposeful at this stage to allow for a wider and quicker exploration (and elimination) of available technical solutions, in the future, it is essential that a larger sample is tested including populations that have rehabilitation or other health-related need to engage with gait training e.g. athletes, stroke patients. Additionally, conducting extensive testing with health professionals in situ to gather feedback on the system's clinical delivery, adoption modes and potential commercialisation value.

In the future, additional potential advantages of AR/VR for gait training and rehabilitation could include incorporating verbal guidance from a remote personal trainer, who would provide real-time instructions and corrections during the training sessions. Finally, expanding the system's capabilities to support remote multi-user sessions could introduce a community aspect to rehabilitation, allowing patients to engage in training together.

7 Conclusion

We presented XRGait, a novel gait training visualization system that integrates wearable sensors – pressure insoles and IMU sensors, with immersive technologies to enhance real-time gait training and rehabilitation. The system provides real-time visual feedback through SensorDot, HeatMap, and full lower-body motion data and was evaluated across mobile AR, VR, and non-VR platforms.

The results show preference for the VR gait training environments that were found to have the potential to enhance embodiment, engagement and exercise adherence. HeatMap and the lower body visualisations were also preferred over SensorDot for their intuitive and easy-to-understand nature, which provided clear, actionable feedback for posture correction. Participants expressed interest in personalizing the VR experience and integrating these systems with existing health apps to further motivate and improve rehabilitation regimens.

In summary, XRGait offers a promising direction for future gait training interventions and provides a foundation for designing more effective VR and AR rehabilitation systems that align with user preferences, potentially enhancing gait training outcomes.

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